

Lab Session : Stabilization of a Segway System ¹

We are interested in the problem of stabilization of a Segway system around the vertical position. We consider the a simplified model of the system such as illustrated if Figure 1 : a mass m_p connected to a wagon of mass m_c via a massless arm of length L . The position of the center of the body is p , its speed is $v = \dot{p}$ and the angle of the arm with the vertical line is θ . The forces applied on the system are : u the force generated by the electrical motor, the viscous friction forces determined by the coefficients d_1 fo the linear movement of the wagon and d_2 for the rotational movement.

Consider the following numerical values :

- $m_c = 1.5 \text{ kg}$
- $m_p = 0.5 \text{ kg}$
- $g = 9.82 \text{ m/s}^2$
- $L = 1 \text{ m}$
- $d_1 = 0.01 \text{ N.s/m}$
- $d_2 = 0.01 \text{ N.s.m}$

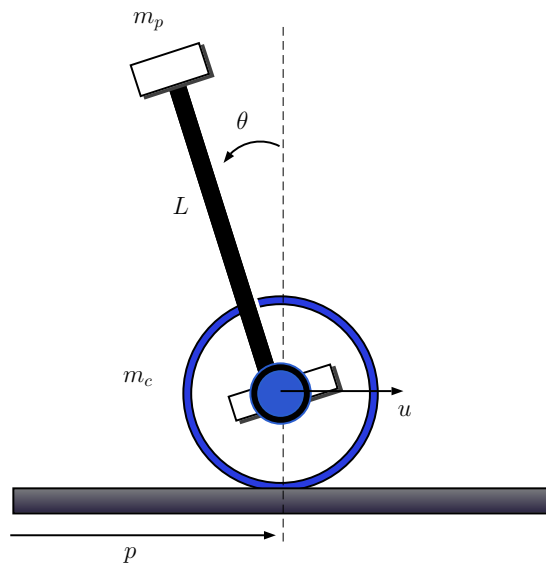


FIGURE 1 – Simplified model of a Segway.

1. Determine the kinetic energy $\mathcal{T}$ and the potential energy $\mathcal{U}$ of the system and find the Lagrangien $\mathcal{L} = \mathcal{T} - \mathcal{U}$.

1. From the [University of Kaiserslautern](#).

Consider the Euler-Lagrange equation :

$$\begin{aligned}\frac{d}{dt}\mathcal{L}_{\dot{p}} - \mathcal{L}_p &= -d_1\dot{p} + u \\ \frac{d}{dt}\mathcal{L}_{\dot{\theta}} - \mathcal{L}_\theta &= -d_2\dot{\theta}\end{aligned}$$

where $\mathcal{L}_z$ is the partial derivative of $\mathcal{L}$ with respect to z . Show that the dynamics of the system are given by :

$$\begin{aligned}(m_p + m_c)\ddot{p} - m_p L\ddot{\theta}\cos(\theta) + m_p L\dot{\theta}^2\sin(\theta) &= -d_1\dot{p} + u \\ m_p L^2\ddot{\theta} - m_p L\ddot{p}\cos(\theta) - m_p g L\sin(\theta) &= -d_2\dot{\theta}\end{aligned}$$

2. Find the equilibrium points of the system.
3. Find a linearized state-space representation of the system around each equilibrium point.

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

Define the state and give the matrices A and B .

4. Design a continuous-time linear state-feedback controller that stabilizes the system around its instable equilibrium.

You can either design a Linear Quadratic Regulator that stabilizes the system and minimizes the following cost

$$J(\mathbf{u}) = \int_{t_0=0}^{\infty} \mathbf{x}^T(t)Q\mathbf{x}(t) + \mathbf{u}^T(t)R\mathbf{u}(t)dt$$

with

$$Q = 2 * I \quad R = 5$$

Or, you can use the pole placement method in order to obtain the following poles in closed-loop : $[-3.4 \quad -3.8 \quad -0.4287 + 0.3627i \quad -0.4287 - 0.3627i]$.

5. **Simulink :** Complete the file *Segway Stabilization init.m*, and the Simulink file *Segway Stabilization.slx* and run the simulation.
6. Repeat the last two questions while considering only θ and p as possible measurements.